

Yoyo's Club - Synthesis Report

What is this project

So basically Yoyo is a nightclub owner but he manage everything on paper. Like staff schedule on loose sheets, salary calculation by hand, and the blacklist only in the bouncer head. So we have to make a proper web dashboard for him so he can manage everything from one place.

What I did

I was assigned the DevWeb module so I build the full admin panel. Here is what I made:

- Login page with email and password and JWT token
- Dashboard with stat cards and 3 charts (hours, payroll, attendance)
- Staff page where Yoyo can hire and fire employees
- Payroll page with salary calculation and Stripe payment
- Tickets page to create events and sell entry tickets

Tech stack

- Frontend: React + Vite, Recharts, Axios
- Backend: FastAPI with Python
- Database: PostgreSQL on Supabase (cloud)
- Auth: JWT token + sha256 password hashing
- Payment: Stripe sandbox

How salary calculation works

I made 3 different formulas based on contract type:

- Full-time: $\text{hours} \times \text{rate} + 10 \text{ percent bonus}$
- Part-time: $\text{hours} \times \text{rate}$ (no change)
- Freelance: $\text{hours} \times \text{rate} \text{ minus } 20 \text{ percent tax}$

Problems I faced

- Supabase direct URL was not resolving on my network so I switched to pooler URL and it worked
- Python 3.14 has issue with bcrypt so I switched to sha256_crypt
- I had duplicate route in tickets.py so Python was using wrong version, I fixed by rewriting the file clean
- Attendance was adding old data to new so I added reset logic when new event is created

Design pattern

I follow MVC pattern. Models are SQLAlchemy tables, Controllers are FastAPI routes, Views are React pages. All API calls are in one file `api.js` so easy to maintain. CSS is separate file per page.